STAT 201A: Homework 2

Ian Abrash

October 24, 2026

Problem 1. A closet contains n pairs of shoes. If $2r$ shoes are chosen at random ($2r < n$), what is the probability that there will be no matching pair in the sample?

Solution:

There are $\binom{2n}{2r}$ equally likely subsets of $2r$ shoes. To have no matching pair, choose $2r$ different pairs from the n pairs, then choose one of the two shoes from each selected pair. Thus there are $\binom{n}{2r}2^{2r}$ favorable subsets, and

$$P(\text{no matching pair}) = \frac{\binom{n}{2r}2^{2r}}{\binom{2n}{2r}}.$$

Problem 2. Suppose that 5% of men and .25% of women are color-blind. A person is chosen at random and that person is color-blind. What is the probability that the person is male? (Assume males and females to be in equal numbers.)

Solution:

Let M denote male, M^c denote female, and C denote color-blind. By the Law of total probability,

$$P(C) = P(C \mid M)P(M) + P(C \mid W)P(W) = 0.05(0.5) + 0.0025(0.5) = 0.02625.$$

Bayes' rule gives us

$$P(M \mid C) = \frac{0.05(0.5)}{0.02625} = \frac{20}{21} \approx 0.95238.$$

Problem 3. Prove each of the following statements. (Assume that any conditioning event has positive probability.)

(a) If $P(B) = 1$, then $P(A|B) = P(A)$ for any A

(b) If $A \subset B$, then $P(B|A) = 1$ and $P(A|B) = P(A)/P(B)$

(c) If A and B are mutually exclusive, then

$$P(A|A \cup B) = \frac{P(A)}{P(A) + P(B)}$$

(d) $P(A \cap B \cap C) = P(A|B \cap C)P(B|C)P(C)$

Solution:

(a) Since $P(B^c) = 0$ and $A \cap B^c \subseteq B^c$, gives us $P(A \cap B^c) = 0$. Which leads us to $A = (A \cap B) \cup (A \cap B^c)$ implies $P(A \cap B) = P(A)$. Therefore

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = P(A).$$

(b) If $A \subset B$, then $A \cap B = A$. Hence,

$$P(B \mid A) = \frac{P(A)}{P(A)} = 1, \qquad P(A \mid B) = \frac{P(A)}{P(B)}.$$

(c) By the definition of conditional probability,

$$P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)}$$

By Distributive Law,

$$A \cap (A \cup B) = (A \cap A) \cup (A \cap B)$$

By definition of mutual exclusivity,

$$A \cap B = \emptyset; A \cap (A \cup B) = A \cup \emptyset = A$$

By theorem 1.2.9,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B)$$

Then,

$$P(A \mid A \cup B) = \frac{P(A)}{P(A) + P(B)}.$$

(d) By the associative rule and multiplication rule,

$$\begin{aligned} P(A \cap B \cap C) &= P(A \cap (B \cap C)) \\ P(A \cap (B \cap C)) &= P(A|B \cap C)P(B \cap C) \\ P(B \cap C) &= P(B|C)P(C) \\ P(A \cap B \cap C) &= P(A|B \cap C)P(B|C)P(C) \end{aligned}$$

Problem 4. A pair of events A and B cannot be simultaneously *mutually exclusive* and *independent*. Prove that if $P(A) > 0$ and $P(B) > 0$, then:

(a) If A and B are mutually exclusive, they cannot be independent.

(b) If A and B are independent, they cannot be mutually exclusive.

Solution:

(a) If A and B are mutually exclusive, then $A \cap B = \varnothing$, so $P(A \cap B) = 0$. But $P(A)P(B) > 0$, so $P(A \cap B) \neq P(A)P(B)$. So, they are not independent.

(b) If A and B are independent, then $P(A \cap B) = P(A)P(B) > 0$. However, $A \cap B$ is nonempty, so they are not mutually exclusive.

Problem 5. A disease has prevalence 1%. A screening test is positive with probability 0.95 for a person with the disease and with probability 0.10 for a person without it. A randomly chosen person takes the test twice. Let $D =$ "the person has the disease" and $T_i =$ "test i is positive" $i = 1, 2$. Assume the two results are independent **given** the disease status:

$$P(T_1 \cap T_2|D) = P(T_1|D)P(T_2|D), \text{ and the same with } D^c \text{ in place of } D$$

(a) Find $P(D|T_1)$

(b) Find $P(D|T_1 \cap T_2)$.

(c) Show that you get the same answer as (b) by applying Bayes' rule to test 2 alone, using your answer to (a) as the "prior" probability of D. Explain why this works.

(d) Are T_1 and T_2 independent? Compute $P(T_1 \cap T_2)$ and $P(T_1)P(T_2)$, and explain in words why "independent given D" does not imply "independent."

Solution:

(a) By law total probability,

$$P(T_1) = 0.95(0.01) + 0.10(0.99) = 0.1085.$$

Therefore by definition of conditional probability,

$$P(D \mid T_1) = \frac{0.95(0.01)}{0.1085} = \frac{19}{217} \approx 0.08756.$$

(b) Conditional independence and the law of total probability give us

$$P(T_1 \cap T_2) = (0.95)^2(0.01) + (0.10)^2(0.99) = 0.018925.$$

Thus by definitiion of conditional probability,

$$P(D \mid T_1 \cap T_2) = \frac{(0.95)^2(0.01)}{0.018925} = \frac{361}{757} \approx 0.47688.$$

(c) Write $p = P(D \mid T_1) = 19/217$. Conditional independence implies $P(T_2 \mid D, T_1) = 0.95$ and $P(T_2 \mid D^c, T_1) = 0.10$. Applying Bayes' rule within the population with T_1 positive gives

$$P(D \mid T_1, T_2) = \frac{0.95p}{0.95p + 0.10(1-p)} = \frac{0.95(19)}{0.95(19) + 0.10(198)} = \frac{361}{757}.$$

(d) Both tests have marginal positive probability 0.1085, so

$$P(T_1)P(T_2) = (0.1085)^2 = 0.01177225 \neq 0.018925 = P(T_1 \cap T_2).$$

They are therefore not independent. Both results depend on the same disease status. A first positive result increases the probability of disease, which in turn increases the chance that the second result is positive.

Problem 6. Prove that the following functions are cdfs.

(a) $\frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x), \quad x \in (-\infty, \infty)$

(b) $e^{-e^{-x}}, \quad x \in (-\infty, \infty)$

(c) the function defined in (1.5.6)

For each, state whether the corresponding random variable is continuous, discrete, or neither (Definition 1.5.7).

Solution:

A function is a cdf if it is nondecreasing, right-continuous, and has limits 0 and 1 at $-\infty$ and $+\infty$, respectively.

(a) For $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$,

$$F'(x) = \frac{1}{\pi(1+x^2)} > 0.$$

Thus F is increasing and continuous, hence right-continuous. Since $\arctan x \to \pm\pi/2$ as $x \to \pm\infty$, the required limits are 0 and 1. The corresponding random variable is continuous.

$$F'(x) = e^{-x}e^{-e^{-x}} > 0.$$

It is increasing and continuous. As $x \to -\infty$, $e^{-x} \to \infty$ and $F(x) \to 0$; as $x \to \infty$, $e^{-x} \to 0$ and $F(x) \to 1$. The corresponding random variable is continuous, with density $f(x) = e^{-x}e^{-e^{-x}}$.

(b) From (1.5.6), for $0 < \epsilon < 1$,

$$F_Y(y) = \begin{cases} \dfrac{1-\epsilon}{1+e^{-y}}, & y < 0, \\ \epsilon + \dfrac{1-\epsilon}{1+e^{-y}}, & y \geq 0. \end{cases}$$

Away from zero, the derivative is $(1-\epsilon)e^{-y}/(1+e^{-y})^2 > 0$. At zero,

$$F_Y(0^-) = \frac{1-\epsilon}{2}, \qquad F_Y(0) = \frac{1+\epsilon}{2},$$

so the jump is upward, of size ϵ. Thus F_Y is nondecreasing. Both pieces are continuous on their intervals and the right-hand limit at zero equals $F_Y(0)$, so it is right-continuous everywhere. The limits at $-\infty$ and ∞ are 0 and 1. Hence it is a cdf.

The distribution is not continuous or discrete as, $P(Y = 0) = \epsilon > 0$ and a continuous component of total probability $1 - \epsilon > 0$.

Problem 7 (A cdf with jumps)**.** Let X have cdf

$$F(x) = \begin{cases} 0, & x < 0, \\ x/4, & 0 \le x < 1, \\ 1/2, & 1 \le x < 2, \\ 1 - \frac{1}{4}e^{-(x-2)}, & x \ge 2. \end{cases}$$

Write $F(a^-) = \lim_{x \uparrow a} F(x)$.

(a) For any cdf and $a < b$, show that $P(a < X \le b) = F(b) - F(a)$ and $P(X = a) = F(a) - F(a^-)$.

(b) Check the three cdf conditions. Is X continuous, discrete, or neither?

(c) Find $P(X = 1)$, $P(X = 2)$, $P(\frac{1}{2} < X < 2)$, $P(X > 3)$, and $P(X \ge 1 \mid X \le 2)$.

(d) Describe a two-stage experiment: flip a fair coin, then draw from a discrete or a continuous distribution. Specify both distributions.

Solution:

(a) The disjoint decomposition $\{X \le b\} = \{X \le a\} \cup \{a < X \le b\}$ implies

$$P(a < X \le b) = F(b) - F(a).$$

Let $A_n = \{X \le a - 1/n\}$. These events increase to $\{X < a\}$, so continuity of probability from below gives

$$P(X < a) = \lim_{n \to \infty} F(a - 1/n) = F(a^-).$$

Since $\{X \le a\} = \{X < a\} \cup \{X = a\}$ is a disjoint union, $P(X = a) = F(a) - F(a^-)$.

(b) Each piece is nondecreasing. The boundary values are

$$F(0^-) = F(0) = 0, \quad F(1^-) = \tfrac{1}{4} < \tfrac{1}{2} = F(1), \quad F(2^-) = \tfrac{1}{2} < \tfrac{3}{4} = F(2).$$

The jumps are upward, so F is nondecreasing on all of $\mathbb{R}$. Each defining piece is continuous from the right at its included endpoint, so F is right-continuous. Also $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$. Thus F is a cdf.

$$\begin{aligned} P(X = 1) &= \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4}, \\ P(X = 2) &= \tfrac{3}{4} - \tfrac{1}{2} = \tfrac{1}{4}, \\ P(\tfrac{1}{2} < X < 2) &= F(2^-) - F(\tfrac{1}{2}) = \tfrac{1}{2} - \tfrac{1}{8} = \tfrac{3}{8}, \\ P(X > 3) &= 1 - F(3) = \frac{1}{4e}, \\ P(X \ge 1 \mid X \le 2) &= \frac{F(2) - F(1^-)}{F(2)} = \frac{3/4 - 1/4}{3/4} = \tfrac{2}{3}. \end{aligned}$$

(c) On heads, draw from the discrete distribution assigning probability 1/2 each to 1 and 2. On tails, draw from the continuous distribution with density

$$g(x) = \begin{cases} 1/2, & 0 < x < 1, \\ \frac{1}{2}e^{-(x-2)}, & x > 2, \\ 0, & \text{otherwise.} \end{cases}$$

This density integrates to $1/2 + 1/2 = 1$. Its cdf is therefore $x/4$ on $[0, 1)$, $1/2$ on $[1, 2)$, and $3/4 + \frac{1}{4}(1 - e^{-(x-2)}) = 1 - \frac{1}{4}e^{-(x-2)}$ on $[2, \infty)$, with value zero below zero, exactly as required.

Problem 8. An electronic device has lifetime T with pdf $f_T(t) = \frac{1}{1.5}e^{-t/1.5}$ for $t > 0$, and zero otherwise. Its value is $V = 5$ if it fails before $t = 3$; otherwise $V = 2T$. Find the cdf of V.

Solution: Integrating the density gives $P(T \le t) = 1 - e^{-t/1.5}$ for $t > 0$. If $T < 3$, then $V = 5$; if $T \ge 3$, then $V \ge 6$. Thus $F_V(v) = 0$ for $v < 5$, while for $5 \le v < 6$,

$$F_V(v) = P(T < 3) = 1 - e^{-2}.$$

For $v \ge 6$, the event $\{V \le v\}$ is the disjoint union of $\{T < 3\}$ and $\{3 \le T \le v/2\}$, which together equal $\{T \le v/2\}$. Consequently

$$F_V(v) = \begin{cases} 0, & v < 5, \\ 1 - e^{-2}, & 5 \le v < 6, \\ 1 - e^{-v/3}, & v \ge 6. \end{cases}$$

Problem 9. A pmf and a pdf with the same tail

(a) Let $f_X(k) = c/[k(k+1)]$ for $k = 1, 2, \ldots$, and zero otherwise. Find the normalizing constant c.

(b) Find $F_X(x)$ for every real x, sketch it, and find $P(X > n)$ for $n = 1, 2, \ldots$.

(c) Let $f_Y(y) = c/y^2$ for $y > 1$, and zero otherwise. Find c, $F_Y(y)$, and $P(Y \geq n)$.

(d) Show that $\lfloor Y \rfloor$ has the same distribution as X.

(e) Explain why a pmf always satisfies $f_X(x) \leq 1$. Must a pdf satisfy $f_Y(y) \leq 1$? Prove it or give a counterexample.

Solution:

(a) Since $1/[k(k+1)] = 1/k - 1/(k+1)$, the sum telescopes:

$$\sum_{k=1}^{m} \frac{c}{k(k+1)} = c\left(1 - \frac{1}{m+1}\right) \longrightarrow c.$$

The probabilities must sum to one, so $c = 1$. All terms are nonnegative, hence this choice defines a pmf.

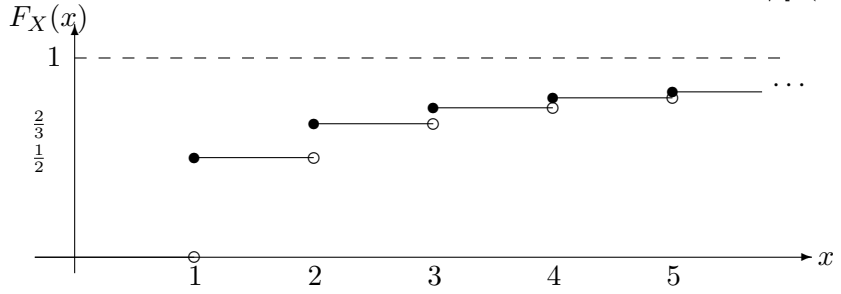
(b) For $x < 1$, $F_X(x) = 0$. For $x \geq 1$, let $m = \lfloor x \rfloor$. Then

$$F_X(x) = \sum_{k=1}^{m} \frac{1}{k(k+1)} = 1 - \frac{1}{m+1}.$$

Therefore

$$F_X(x) = \begin{cases} 0, & x < 1, \\ 1 - \dfrac{1}{\lfloor x \rfloor + 1}, & x \geq 1. \end{cases} \qquad P(X > n) = 1 - F_X(n) = \frac{1}{n+1}.$$

The graph is a right-continuous step function: it equals $1/2$ on $[1, 2)$, $2/3$ on $[2, 3)$, $3/4$ on $[3, 4)$, and so on, approaching 1. Each jump at integer $k \geq 1$ has height $1/[k(k+1)]$.



(c) Normalization requires

$$1 = \int_1^\infty \frac{c}{y^2}\, dy = c,$$

so $c = 1$. Integrating up to y gives

$$F_Y(y) = \begin{cases} 0, & y \le 1, \\ 1 - \dfrac{1}{y}, & y > 1. \end{cases}$$

Since Y is continuous, for every integer $n \ge 1$,

$$P(Y \ge n) = \frac{1}{n}.$$

(d) For integer $k \ge 1$,

$$P(\lfloor Y \rfloor = k) = P(k \le Y < k+1) = \int_k^{k+1} \frac{1}{y^2}\, dy = \frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}.$$

(e) A pmf is a point probability: $f_X(x) = P(X = x) \le P(\Omega) = 1$ by monotonicity. A pdf is a density, so only its integrals over events must be probabilities. For example,

$$h(y) = \begin{cases} 2, & 0 < y < 1/2, \\ 0, & \text{otherwise} \end{cases}$$

is nonnegative and integrates to $2(1/2) = 1$, hence is a valid pdf, even though $h(y) = 2 > 1$ throughout $(0, 1/2)$.